Pre-processing of Input Sequence

Sequence of words with Length N

Word 0 | Word 1 | Word 2 | Word 3 | Word 4 | Word 5 | ... | Word $_{N-2}$ | Word $_{N-1}$

Word Embedding

e.g. word2vec

X_0 X_1 X_2 X_3 X_4 X_5 ... X_{N-2} X_{N-1}

Position Information

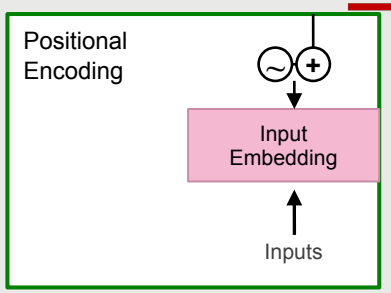
Pos 0 | Pos 1 | Pos 2 | Pos 3 | Pos 4 | Pos 5 | ... | Pos $_{N-2}$ | Pos $_{N-1}$

$+\delta$ $+\delta$ $+\delta$ $+\delta$ $+\delta$ $+\delta$

Final Embedding

Combine Word Encoding and Position Information (for Each Word)

I/P 0 | I/P 1 | I/P 2 | I/P 3 | I/P 4 | I/P 5 | ... | I/P $_{N-2}$ | I/P $_{N-1}$



Input to Encoder

Input to first Self Attention Layer in Transformer